Financial System Integration and Shock Transmission

1MT404 Financial Stability

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Agenda

1. The Changing Financial Landscape
2. Network and Linkages: Banks vs. NBFI
3. The Rise of Private Credit
4. Channels of Shock Transmission
5. Historical Episodes & Lessons
6. Vulnerabilities & Policy Implications
7. Conclusion

The Great Migration: From Banks to Non-Banks

- **Historical Pivot:** Since the Global Financial Crisis (2008), the financial system has undergone a "de-banking" process. While banks were constrained by strict regulation (Basel III), financial activity migrated to the less regulated NBFI sector.
- **Total Assets:** Global financial assets reached approximately **$500 trillion** in 2024.
- **The 51% Milestone:** The NBFI sector now manages **$256.8 trillion**, representing more than half (51%) of all global financial assets – *FSB 2025*.
- **Composition:** Investment funds (mutual, hedge), pension funds, and insurance companies play a dominant role.

The "New Normal" for Financial Stability

Financial stability no longer depends solely on the solvency of banks, but on **market liquidity** and the resilience of investment funds.

Structural Drivers of the NBFI Expansion

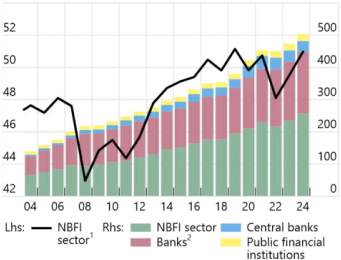
Why has the NBFI sector grown so aggressively?

1. **Regulatory Tightening (Push Factor):** Strict capital requirements on banks pushed riskier activities (e.g., mid-market lending) off bank balance sheets.
2. **Search for Yield (Pull Factor):** The long period of low interest rates forced investors to seek returns in illiquid and alternative assets (Private Credit, Private Equity).
3. **Technology & Institutionalization:** Digitalization enabled lightning-fast capital moves and the rise of giant asset managers (BlackRock, Vanguard) with market power exceeding many central banks.
4. **Demographics:** Aging populations in developed countries increase assets in pension funds and life insurance.

NBFI Assets as % of Total (FSB 2025)

Total global financial assets

Share of total financial assets — USD trillion



Composition of the global financial system[4]

	Total global financial assets	Central banks	Banks[2]	PFIs[3]	NBFI sector
Size at end-2024 (USD trillion)	503.7	34.1	191.3	21.5	256.8
Share of total global financial assets (%)	100.0	6.8	38.0	4.3	51.0
Growth in 2024 (year-over-year, %)	6.3	-4.2	4.7	3.1	9.4
Growth 2018–23 (annualised growth, %)	6.1	6.6	6.3	5.2	5.9

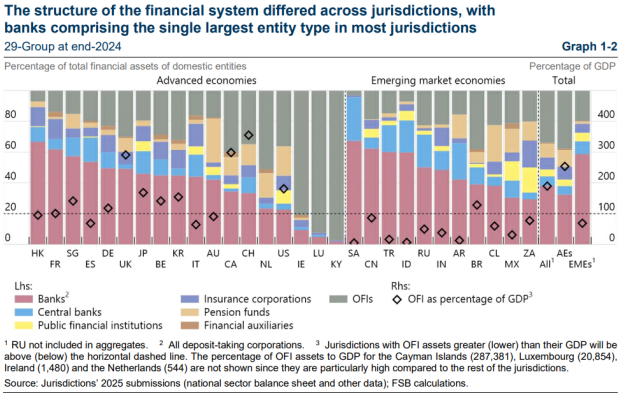
[1] NBFI includes ICs, PFs, OFIs, and financial auxiliaries. [2] All deposit-taking corporations. [3] Public financial institutions.

Source: Jurisdictions' 2025 submissions (national sector balance sheet and other data) and 2021 submission for Russia; FSB calculations.

Takeaway: Non-banks now hold just over half of global financial assets ($257 trillion of $504 trillion at end-2024) and grew twice as fast as banks in 2024 (9.4% vs 4.7%).

Source: FSB, Global Monitoring Report on Nonbank Financial Intermediation 2025 (December 2025), Graph 1-1.

NBFI Assets as % of Total by Country (FSB 2025)



Takeaway: Banks are still the largest sector in most of the 29 jurisdictions; the global NBFI weight comes from the US and from financial centres such as the Cayman Islands, Luxembourg and Ireland, where OFI assets are many times GDP.

Source: FSB, Global Monitoring Report on Nonbank Financial Intermediation 2025 (December 2025), Graph 1-2.

Anatomy of the NBFI Sector: The "Narrow Measure"

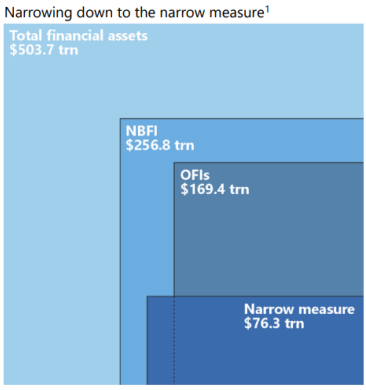
NBFI is not a monolith. Regulators (FSB) focus on the **Narrow Measure** ($76.3 trillion), consisting of entities that may pose systemic risk:

- **EF1: Investment Funds** ($55T) – Vulnerable to liquidity mismatch (daily redemptions vs. illiquid assets).
- **EF2: Finance Companies** ($4T) – Dependent on short-term wholesale funding, providing credit.
- **EF3: Broker-Dealers** ($9T) – Key for market-making, highly interconnected through repo markets.
- **EF4: Insurance-led & Credit Enhancement** ($0.5T) – Providing guarantees.
- **EF5: Securitization Vehicles** ($7T) – Transforming risky loans into securities.

Focus 2025–2026

The fastest-growing component within EF1 is now **Private Credit**, which is becoming a direct competitor to bank loans.

The "Narrow Measure" (2025)



Takeaway: Of the $257 trillion NBFI sector, $76 trillion is the narrow measure – activities that can create bank-like risks (liquidity and maturity transformation, leverage) – and it grew 12% in 2024.

Source: FSB, Global Monitoring Report on Nonbank Financial Intermediation 2025 (December 2025), Graph 0-1.

The Local Context: Investment Funds (CNB Perspective)

While global systemic risk often focuses on mega-funds, local markets are shaped by standard investment funds.

- **Retail vs. Qualified Investors:** Shift towards alternative investment funds (AIFs) alongside traditional UCITS.
- **Liquidity Transformation:** Funds offer daily liquidity to retail investors while holding less liquid assets (e.g., real estate, corporate bonds).
- **Systemic footprint:** Even without massive leverage, herding behavior among retail investors can cause local market disruptions.

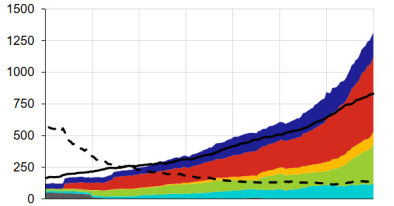
CNB's View on Vulnerabilities

"Investment funds democratize capital markets, but their promise of daily liquidity can become a systemic vulnerability during market stress if underlying assets cannot be quickly liquidated."

– Adapted from CNB Blog (Everything you wanted to know about funds)

Investment Fund Assets in the Czech Republic

Graf 6 – Aktiva segmentu českých investičních fondů
(v mld. Kč; počet fondů v jednotkách, HHI v rozmezí od 0 až 10 000)

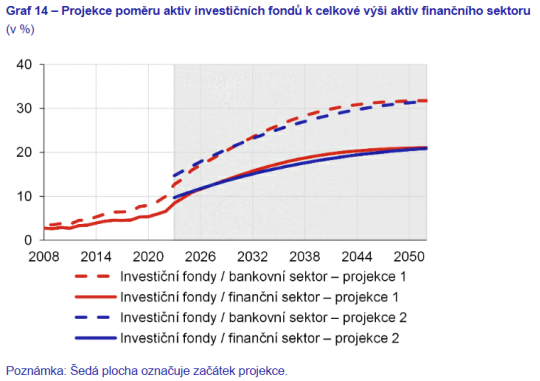


Poznámka: Počet zahrnuje i podfondy. HHI značí Herfindahl – Hirschmanův index měření koncentrace. Fondy kolektivního investování jsou rozděleny dle převažujícího typu aktiv.

Takeaway: Czech investment fund assets grew roughly tenfold since 2009, to about CZK 1.3 trillion at end-2023, with qualified-investor funds the largest block; the number of funds rose above 800 while concentration (HHI) fell.

Source: ČNB; Časta, Siuda and Holub (2024), "Vše, co jste kdy chtěli vědět o investičních fondech (a báli jste se zeptat)", ČNB blog, 18 September, Graph 6.

Future Projections for the Czech Market



Takeaway: On ČNB projections, investment funds roughly double their weight in the Czech financial system – to about a fifth of financial-sector assets, and about a third of banking-sector assets, by 2050.

Source: Časta, Siuda and Holub (2024), "Vše, co jste kdy chtěli vědět o investičních fondech (a báli jste se zeptat)", ČNB blog, 18 September, Graph 14 (ČNB projections; grey area = projection).

The Wealth Shift: Czech Household Portfolios (CNB Blog)

- **The Migration:** Czech households are moving away from traditional overnight deposits toward **Investment Funds** and **Equities**.
- **Drivers:** Low real interest rates (historically) and a maturing capital market have pushed retail investors into the NBFI sector.
- **The Risk Profile:**
  - **Market Risk Exposure:** Household wealth is now directly tied to stock and bond market volatility.
  - **Liquidity Expectation:** Retail investors often expect "instant" liquidity from funds that may hold illiquid underlying assets.

The "Wealth Effect" Channel

A sharp correction in NBFI asset prices now has a faster, more direct impact on **domestic consumption** (as households feel "poorer" immediately), creating a new macro-prudential concern for the CNB.

The Hybrid Reality: Banks as Infrastructure Providers

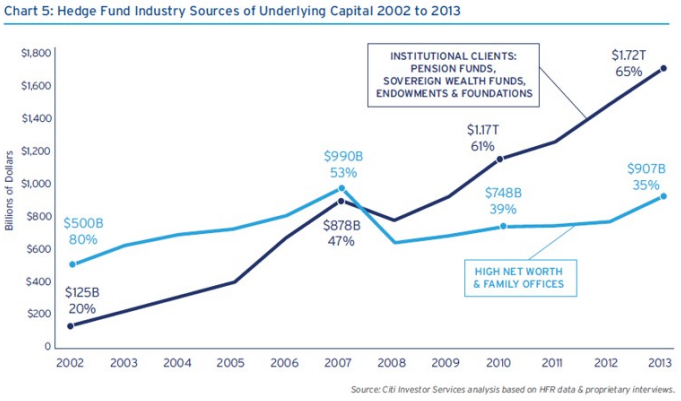
The boundary between banks and NBFIs is blurring. Banks are no longer just "lenders" but have become **infrastructure providers** for NBFIs:

- **Prime Brokerage:** Lending securities and providing leverage (Hedge Funds).
- **Custody & Settlement:** Holding assets and clearing trades.
- **Liquidity Backstops:** Providing RCFs (Revolving Credit Facilities).

Systemic Implication

Banks are now **indirectly exposed** to risks they cannot fully control (e.g., private equity strategies or highly leveraged crypto-proxies).

Hedge Funds - Investors



Takeaway: Institutions' share of hedge fund capital rose from about a fifth (2002) to about two-thirds (2013): more stable money, but it ties pension savings to hedge fund leverage.

Source: Citi Investor Services analysis based on HFR data and proprietary interviews.

Compensation and Incentive Structure

- Management fee: 1.5–2.0% p.a. of AUM
- Performance fee: 15–20% of return above „hurdle rate“ (usually 0–8%)
- High-water mark: performance fee only paid on new peak value
- Lock-up periods: 6–24 months (limits capital outflow during losses)
- Incentive alignment: Managers often invest own capital („skin in the game“)

This structure supports high risk-taking and search for alpha but increases procyclicality during large drawdowns.

Hedge Fund Returns: Correlations by Strategy (1990–2014)

Figure 1: Correlation table comparing stocks, bonds and hedge fund sub-sectors

From Jan. 1990 to Jan. 2014		1	2	3	4	5	6	7	8
Developed Market Equities	1	1.00							
Developed Government Bonds	2	-0.01	1.00						
Event-Driven Index	3	0.61	-0.09	1.00					
Equity Hedge Index	4	0.63	-0.05	0.84	1.00				
Equity Market Neutral Index	5	0.16	0.04	0.42	0.51	1.00			
Quantitative Directional	6	0.71	-0.02	0.82	0.89	0.34	1.00		
Macro Index	7	0.34	0.35	0.51	0.55	0.34	0.55	1.00	
Relative Value Index	8	0.41	-0.05	0.77	0.68	0.41	0.57	0.34	1.00

Source: Bloomberg, HFRI, Barclays.

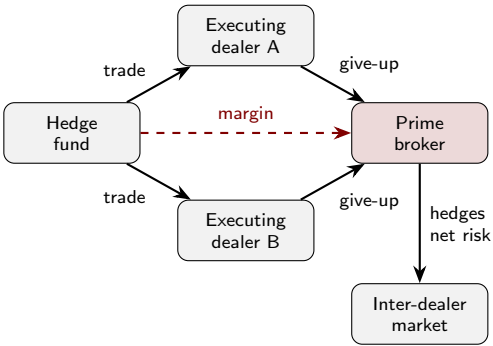
- Monthly HFRI strategy indices vs developed-market equities and government bonds, Jan 1990 – Jan 2014
- **Equity beta:** quantitative directional 0.71, equity hedge 0.63, event-driven 0.61 with equities
- **Diversifiers:** equity market neutral 0.16, macro 0.34; macro is the only strategy correlated with bonds (0.35)
- **Near-duplicates:** equity hedge vs quantitative directional 0.89, event-driven vs equity hedge 0.84
- **Caveats:** full-sample averages – correlations rise in crises (2008); self-reported index data understate risk (survivorship, smoothed returns)

Takeaway: Most strategies are leveraged equity bets that move together, so losses – and the margin calls that follow – hit many funds and their prime brokers at the same time.

Source: Bloomberg, HFRI, Barclays.

How Prime Brokerage Works

- **Prime broker (PB)**: a large bank, usually a G-SIB, offers hedge funds a bundle – margin financing, securities lending for short sales, synthetic exposure via total return swaps, custody, clearing and settlement
- The fund posts **margin**; the PB sets haircuts and limits – its **credit line** is the fund's binding constraint
- In FX and OTC derivatives: **give-up** arrangements – the fund trades with many executing dealers, the trades are booked with the PB, which faces the market (spot, forwards, NDFs, options)
- Margin is a small fraction of notional $\Rightarrow$ high **embedded leverage**
- After large losses (SNB 2015, Archegos 2021) PBs cut lines across *all* clients $\Rightarrow$ forced deleveraging (substitutability risk)



The Prime Brokerage Service Bundle

Service	What the fund gets	Risk for the prime broker
Margin lending	Cash to buy long positions, secured by those positions	Credit risk if collateral falls faster than margin can be called (gap risk)
Securities lending	Borrowed securities to sell short (sourced from agent lenders or other clients)	Recall risk, short squeezes
Synthetic financing (TRS, CFDs)	Economic exposure without owning the asset; the PB holds the hedge	Concentration and wrong-way risk; aggregate leverage invisible across PBs (Archegos)
Repo financing	Funding of bond positions, e.g. the Treasury cash–futures basis trade	Collateral values; haircuts often thin in bilateral repo
FX and derivatives give-up	One credit relationship, access to many executing dealers	PB guarantees the fund's trades to dealers within credit limits
Custody, clearing, reporting, capital introduction	Operations outsourced, introductions to investors	Operational and reputational risk

PB revenue: financing spreads over the benchmark rate, securities-lending fees (above all for hard-to-borrow stocks), swap spreads and commissions – so the business grows with client leverage.

How Prime Brokers Fund Themselves: Collateral Reuse and Internalisation

- **Rehypothecation:** securities that clients post as collateral are re-pledged by the PB to raise its own funding (repo, securities lending)
  - US: reuse capped at 140% of the client's debit balance (SEC Rule 15c3-3)
  - UK: no statutory cap – the limit is whatever the contract says (Singh & Aitken 2010)
- **Internalisation:** one client's long positions fund another client's shorts inside the PB – cheaper and balance-sheet light, but only as stable as the client mix
- **Collateral velocity:** the same security supports several layers of funding – counting reuse, the shadow banking system was at least 50% larger than previously measured (Singh & Aitken 2010)
- **Lehman 2008:** hedge funds' rehypothecated assets at the London broker-dealer (LBIE) were caught in the insolvency – clients became unsecured creditors $\Rightarrow$ since then: several PBs per fund ("multi-prime"), limits on reuse, segregated custody

Why it matters

Collateral reuse makes funding cheap in good times and fragile in bad times: when clients pull their collateral, the PB loses funding at the same moment.

Margin Terms and Leverage: A Worked Example

Maximum leverage $= 1/m$ (margin per unit of exposure)

Margin regime	m	Leverage
US Reg T, cash equities (rule-based)	50%	2×
Risk-based / portfolio margin, diversified book	≈15%	≈6.7×
Bilateral swaps (illustrative)	10%	10×

Contract terms matter as much as the level: margin **lock-ups** (PB cannot raise margin for a notice period) reduce run risk; **static** margin (fixed share of initial notional) leaves the PB under-collateralised as positions grow.

Fund: capital 100, margin 15% ⇒ position 667

After a 10% price fall	m unchanged	$m \to 25\%$
Position value	600	600
Capital left	33	33
Margin required	90	150
Shortfall (cash to post)	57	117
Forced sale if no new cash	378 (63%)	467 (78%)

A 10% price move wipes out two thirds of capital; a higher margin after the shock forces the fund to sell most of its book – the margin spiral in numbers.

Credit Suisse: a USD 150,000 software fix to raise margins periodically on Archegos swaps was proposed in 2020 but never approved (Risk.net).

Prime Brokers as a Systemic Node: Runs in Both Directions

PB → funds: a funding run

- After losses, or when its own balance sheet is constrained, a PB raises margins and cuts lines for *all* clients ⇒ correlated deleveraging
- The market is concentrated in a few G-SIBs, and exits shrink capacity further:
  - Deutsche Bank's PB clients moved to BNP Paribas (2019, ≈ USD 200 bn of client assets)
  - Credit Suisse exited after Archegos and referred clients to BNP Paribas (November 2021)

Funds → PB: a run on the bank

- When a PB looks weak, hedge funds move cash balances and collateral elsewhere – Bear Stearns (March 2008) and Morgan Stanley (September 2008) lost PB balances fast
- Reused client collateral has to be returned or replaced ⇒ the PB loses funding exactly when it needs it
- Multi-prime makes switching easy – good for funds, faster runs for banks

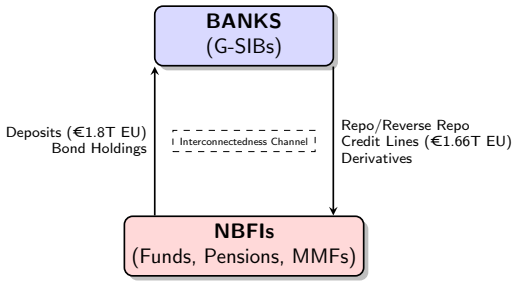
Lesson

Prime brokerage ties hedge fund leverage directly to G-SIB liquidity. Supervisors look at margin practice (dynamic vs static), client concentration and exposures to the same client across PBs – which no single bank can see on its own.

Bank–NBFI Linkages

- **Banks fund NBFIs:** via repo, reverse repo, and committed credit lines (€1.66T in EU).
- **NBFIs fund banks:** via deposits and bank bond holdings (€1.8T in EU).
- **Off-balance-sheet ties:** Derivatives and securitisation pipelines.
- **High Concentration:** Euro-area G-SIBs intermediate 50–80% of derivatives/repos.

Sources: FSB 2025, ECB 2026, IMF GFSR Oct 2025



The Core Nexus: Direct Funding (Balance Sheet)

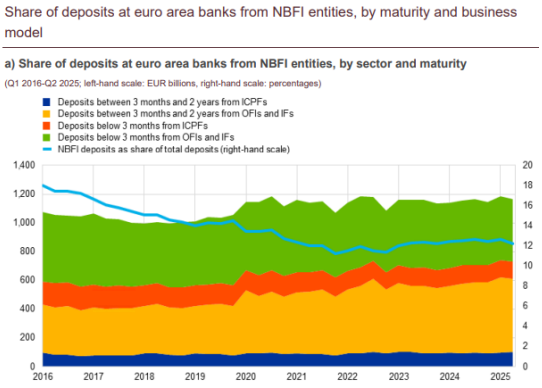
NBFIs Funding Banks (Liabilities Side)

- NBFIs account for roughly **15% of all bank funding** globally.
- In the Euro Area, NBFI deposits in banks amount to **€1.8 trillion** (approx. 12.9% of all deposits) – *ECB 2026*.
- Risk: This wholesale funding is highly runnable. MMFs and investment funds will rapidly withdraw deposits during stress (as seen in March 2020).

Banks Funding NBFIs (Asset Side)

- Banks provide **€1.66 trillion** in direct credit to NBFIs in the EU (10% of total bank assets).
- This includes term loans, revolving credit facilities, and prime brokerage financing for hedge funds and private credit vehicles.
- Risk: Credit risk materialization if leveraged NBFIs default.

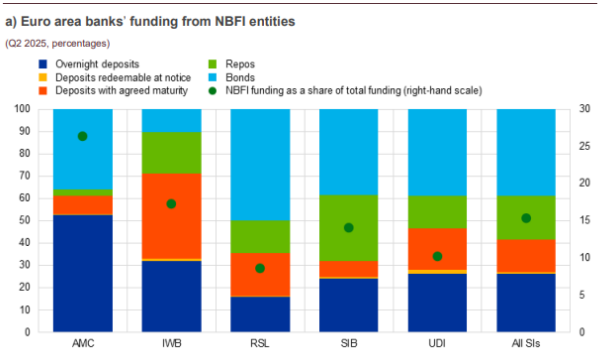
NBFI Deposits in the EA Banking Sector



Takeaway: NBFI entities hold almost EUR 1.2 trillion of euro area bank deposits – about 12% of the total – and close to half of it matures within three months, so this funding can leave quickly.

Source: ESRB and ECB (2026), Financial stability risks from linkages between banks and the non-bank financial intermediation sector, February, Chart 2.1 (panel a).

Bank Funding from NBFIs by Business Model



Takeaway: NBFIs provide about 15% of significant banks' funding – over a quarter at asset managers and custodian banks – and almost half of it is overnight deposits and repos.

AMC = asset managers and custodian banks; IWB = investment and wholesale banks; RSL = retail banks and small lenders; SIB = systemically important banks; UDI = universal and diversified institutions.

Source: ESRB and ECB (2026), Financial stability risks from linkages between banks and the non-bank financial intermediation sector, February, Chart 1 (panel a).

Transmission Channels: ECB 2026 Risk Matrix

The "Tightened Loop" means shocks in the NBFI sector transmit instantly to bank balance sheets.

Risk Factor	NBFI Vulnerability	Bank Impact
Rates ↑	Valuation losses (Bonds)	Margin calls on derivatives
CRE Downturn	Liquidity mismatch (RE Funds)	Credit losses on NBFI loans
Credit Spreads	Forced selling (Fire sales)	MTM losses on trading books

- **The Liquidity Trap:** Margin procyclicality forces NBFIs to sell assets, which lowers collateral values for banks, creating a feedback loop.

Off-Balance Sheet: The Hidden Leverage

- **Repo Markets:** Banks are net borrowers of cash from MMFs and net lenders of cash to hedge funds.
- Reverse repos to hedge funds have **doubled since 2021** (ECB 2026), increasing exposure to highly leveraged players.
- **Derivatives:** NBFIs use banks to access interest rate and FX swaps (notional values exceed $90 trillion globally).
- **Margin Procyclicality:** During market shocks, banks (as clearing members) must issue massive margin calls to NBFIs, forcing NBFI asset fire-sales.

The Archegos Scenario

Synthetic leverage via Total Return Swaps (TRS) allows NBFIs to build massive exposures off-balance sheet. When positions move against them, prime broker banks absorb the ultimate losses.

Indirect Linkages: Portfolio Overlap Channel

Exposures exist even **without direct lending** through common asset holdings.

- **Cosine Similarity Metric:** Measures portfolio overlap. Similarity between Euro-area banks and Pension Funds is > 0.90 for sovereign bonds.
- **Fire-Sale Contagion Loop:**
  1. NBFI faces investor redemptions or margin calls.
  2. NBFI is forced to liquidate highly rated assets (e.g., government bonds, blue-chip corporate debt).
  3. Asset prices plummet (yields spike).
  4. Impact on Banks: Mark-to-market losses on bank trading books, depletion of CET1 capital, and reduced collateral values for bank borrowing.

Step-In Risk: Banks as "Lenders of Last Resort"

Banks often act as the ultimate "lenders of last resort" for the NBFI sector, creating contingent liabilities that only materialize during crises.

Credit Lines & Liquidity Facilities

- Banks provide massive uncommitted and committed credit lines to Private Equity and Private Credit funds.
- In a crisis, funds draw down these lines simultaneously, draining bank liquidity exactly when the bank needs it most.

Reputational Step-In

- Banks sponsor off-balance sheet vehicles (e.g., securitisation SPVs, MMFs).
- Even without a legal obligation, banks may "step in" to rescue a failing affiliated fund to protect their franchise reputation, bringing hidden toxic assets back onto the bank's balance sheet.

Systemic Outlook: Concentration & Stress

- **Extreme Concentration:** The interconnectedness is not evenly distributed. A handful of Global Systemically Important Banks (G-SIBs) intermediate 50–80% of all NBFI derivative and repo transactions.
- **Substitutability Risk:** If one major prime broker steps back, heavily leveraged NBFIs may collapse due to a lack of alternative funding.

IMF 2025 Stress Test Findings

- In a severe NBFI stress scenario, bank CET1 ratios could deplete by an additional **100 basis points** purely due to NBFI contagion.
- **Liquidity drain:** Up to 14% of banks would face negative net liquidity if NBFIs simultaneously withdraw their deposits and max out their credit lines.

Private Credit: The "Shadow" Lending Engine

Private credit has evolved from a niche asset class to a **$2.1 trillion** global market, competing directly with traditional syndicated loans.

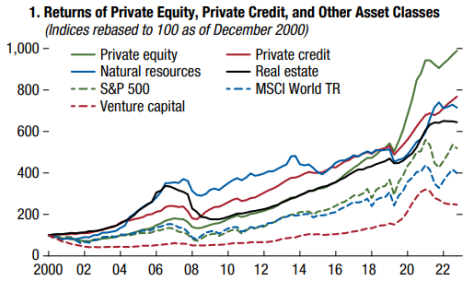
- **The Core Mechanism:** Non-bank lending to corporate borrowers (mid-market) via privately negotiated, non-traded loans.
- **Regulatory Arbitrage:** Stringent Basel III/IV requirements made mid-market lending "capital-expensive" for banks, shifting risk to the NBFI sector.
- **Borrower Appeal:** Speed, confidentiality, and "bespoke" terms compared to public bond markets.
- **Market Dominance:** In the US, private credit now competes directly with the syndicated loan and high-yield bond markets.

The Shift in Risk

Credit risk hasn't disappeared; it has migrated from regulated bank balance sheets to "darker" corners of the financial system with lower transparency.

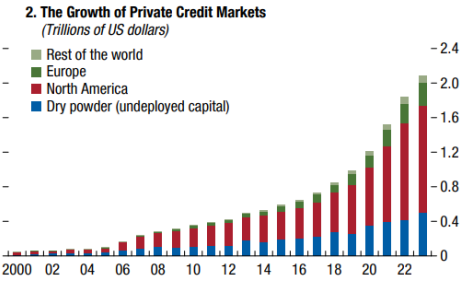
The Exponential Growth of Private Credit

Private credit funds have delivered comparatively higher gross returns ...



Private credit fund managers based in North America manage a material part of the market in other regions.

... and grown exponentially over the past two decades.

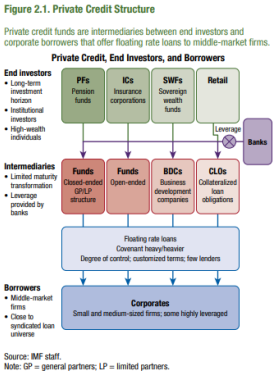


In the United States, private credit size is comparable to leveraged loans and high-yield bond markets.

Takeaway: Private credit has out-returned listed equities since 2000 (gross of fees) and grew to about $2.1 trillion by 2023, mostly in North America; about a quarter of it is committed capital not yet lent ("dry powder").

Source: IMF, Global Financial Stability Report, April 2024, Chapter 2, Figure 2.2 (panels 1–2).

Private Credit Fund Structure



Takeaway: Private credit funds channel long-term money (pension funds, insurers, sovereign wealth funds) to mid-sized, often highly leveraged firms – and banks come back in as lenders to the funds, so their exposure is indirect.

Source: IMF, Global Financial Stability Report, April 2024, Chapter 2, Figure 2.1 (IMF staff).

The Symbiosis: Private Credit and Private Equity (PE)

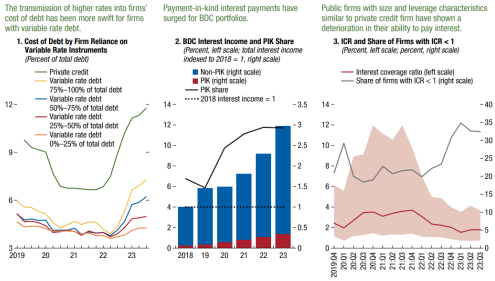
Private credit and Private Equity are two sides of the same coin. According to **IMF GFSR 2024**, over **70% of private credit deals** are "sponsored" by PE firms.

- **The Leveraged Buyout (LBO) Engine:** PE firms use private credit to finance the acquisition of companies. The private credit fund provides the debt, while the PE fund provides the equity.
- **Dry Powder Linkage:** As PE firms raise more capital ("dry powder"), the demand for private credit to leverage those investments grows proportionally.
- **"Extend and Pretend":** In times of stress, PE sponsors and private credit lenders can renegotiate terms privately (e.g., PIK - Payment-in-Kind interest), hiding defaults from the public eye.

Inter-sectoral Contagion

If a large PE sponsor faces a liquidity crisis, it affects not just its equity investors but dozens of private credit funds and the banks providing leverage to those funds.

Structural Vulnerabilities: PIK and Covenants



Sources: BDC 10-K and 10-Q filings; S&P Capital IQ; and IMF staff calculations.
Note: Panel 1 shows the cost of debt, calculated as interest expense divided by total debt. Medians are taken for each bucket of variable rate debt reliance, whereby this reliance is expressed as the ratio of variable rate debt over total debt. The cost of debt within each bucket varies based on credit fundamentals. Private credit rates are based on a sample of BDC portfolios. In panel 2, when interest is paid in kind, no cash flow occurs. Instead, the interest coupon is added—usually at an extra cost—to the loan's principal. Statistics in panel 3 are based on a sample of public firms located in North America with size and leverage characteristics similar to those of borrowers in the private credit universe. It should be noted that interest coverage and debt/EBITDA ratios are usually not reflected in firm-level databases when earnings are negative. This means that the true number of firms with unsustainable interest expense level is (even) higher than indicated by the ICR = 1 threshold. BDC = business development company; EBITDA = earnings before interest, taxes, depreciation, and amortization; ICR = interest coverage ratio; PIK = payment in kind.

- **PIK (Payment-in-Kind):** Allowing borrowers to pay interest with more debt rather than cash – a growing trend in 2025–2026.
- **Covenant-Lite:** Weakening investor protections in exchange for higher yields.

Takeaway: After 2022 private-credit borrowers' cost of debt rose from about 7% to 12%, PIK grew to over a tenth of BDC interest income, and about a third of comparable firms no longer cover interest from earnings (ICR < 1).

Source: IMF, Global Financial Stability Report, April 2024, Chapter 2, Figure 2.4 (BDC 10-K and 10-Q filings; S&P Capital IQ; IMF staff calculations).

Valuation Opacity and "Volatility Smoothing"

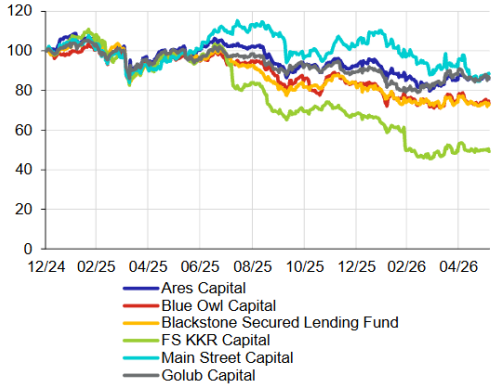
Private markets report lower volatility than public markets—but this is often an illusion created by accounting methods.

- **Mark-to-Model:** Pricing based on internal assumptions rather than market trades.
- **Infrequent Appraisals:** Lags in reflecting the true impact of high interest rates.

Risk Implications

- Underpriced leverage.
- Delayed loss recognition.
- **Sudden "Cliffs":** Risk of sharp, lumpy devaluations when models finally catch up to reality.

Market Prices vs Model Values: Listed Private Credit Funds (BDCs)



Takeaway: Listed BDCs – the only private credit vehicles with a daily market price – trade 12–27% below their end-2024 level, and FS KKR about half: when shares trade below reported NAV, investors are doubting portfolio quality, liquidity or the valuations themselves.

Share prices, index 31 Dec 2024 = 100. BDCs pay high dividends, so total returns fell less than prices.

Source: LSEG; Časta and Maňas (2026), "Za hranicemi veřejných trhů: soukromý kapitál a jeho rizika", ČNB blog, 25 August, Graph 6.

The Leverage Trap: "Leverage on Leverage"

The systemic risk is amplified by layers of debt within the private credit ecosystem.

- **Asset-Level:** Borrowers are already highly indebted (leveraged mid-market firms).
- **Fund-Level:** Private credit funds use **Subscription Lines** (bridge loans from banks) to boost internal rates of return (IRR).

The Macro Concern (IMF 2025)

The sector grew during a "Goldilocks" period of low rates. It remains **untested** in a prolonged high-rate environment with a significant corporate default cycle.

Transmission to the Banking Sector

How does a Private Credit "default wave" hit the regulated banks?

- **Direct Exposure:** Banks provide "subscription lines" (short-term bridge loans) to private credit funds.
- **Asset-Based Finance (ABF):** Banks are increasingly selling their own loan portfolios to private credit funds to clean up balance sheets, creating a feedback loop.
- **Wealth Management:** G-SIBs distribute private credit products to their high-net-worth clients. A collapse in valuations could lead to massive reputational damage and legal liabilities.

"The lack of transparency and high interlinkage with systemic banks make private credit a potential 'blind spot' for financial stability."

— BIS Quarterly Review, 2025

Key Channels of Shock Transmission

1. Liquidity Mismatch

Daily redemptions vs. illiquid underlying assets (Real Estate, Private Credit).

2. Margin & Leverage

Procyclical margin calls and hidden synthetic leverage (Archegos).

3. Interconnected USD Funding

High reliance on FX swaps and global USD liquidity.

4. Correlation & Herding

High portfolio overlap (> 0.9) leads to simultaneous "dashes for cash."

The Shift

Algorithmic trading and digital connectivity mean shocks no longer travel linearly—they propagate exponentially through automated margin processes.

Channel 1: Liquidity Spirals & Fire Sales

This channel transforms an individual fund's liquidity issue into a systemic price collapse.

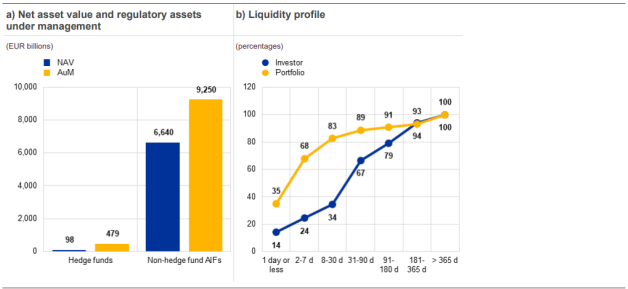
- **The Trigger:** Sudden redemptions or margin calls in open-ended funds.
- **First-Mover Advantage:** Investors exit early to avoid being stuck with illiquid assets, creating a "run" on the fund.
- **The Forced Sale:** Funds sell their most liquid assets (Sovereign bonds) first, depressing prices for all holders (Banks, Pensions).
- **ECB 2026 Observation:** Increased **algorithmic trading** has made these spirals faster and more violent.

March 2020: "Dash for Cash"

Even the US Treasury market—the world's most liquid—faced a shock as NBFIs sold everything to secure cash, requiring massive central bank intervention.

EA Hedge Funds: Liquidity and Leverage

Hedge funds subject to AIFMD rules: net asset value and regulatory assets under management (panel a) and liquidity profile (panel b)



Sources: ECB (AIFMD), NCAs and ESMA.
Notes: Data as at year-end 2024. Panel a): NAV and AuM by AIF type. "Regulatory assets under management" refers to the value of all portfolio assets, including all assets acquired through the use of leverage (borrowing of cash or securities and leverage embedded in derivative positions). Data refer to AIFs of authorised and registered AIFMs excluding feeder funds. Panel b): portfolio and investor liquidity profiles of hedge funds managed and/or marketed by authorised EEA AIFMs. Portfolio profile determined by percentage of the portfolios capable of being liquidated within each specified period; investor profiles depend on the shortest period within which redemption payments could be received. Data refer to AIFs of authorised managers excluding feeder funds and AIFs not reporting liquidity profile data.

Takeaway: Hedge funds under AIFMD hold only EUR 98 billion of equity but about five times that in assets (EUR 479 billion); their portfolios are more liquid than investors' redemption rights, so the risk lies in leverage and margin calls rather than in investor runs.

Source: ECB (AIFMD), NCAs and ESMA; data at end-2024.

Channel 2: Procyclical Margining & Leverage Stack

- **Procyclicality:** Banks/CCPs increase margin requirements (IM/VM) exactly when volatility spikes, draining NBFI cash precisely when it is most scarce.
- **The Leverage Stack:** Leverage is now multi-layered:
  1. **Asset-Level:** Highly indebted corporate borrowers (LBOs).
  2. **Fund-Level:** Subscription lines and repo borrowing.
  3. **Synthetic:** Off-balance sheet Total Return Swaps (TRS).

Hidden Risk: The Archegos Case

Synthetic leverage via swaps allowed one entity to build massive exposures across multiple banks; none of the banks saw the "total" risk until the margin call.

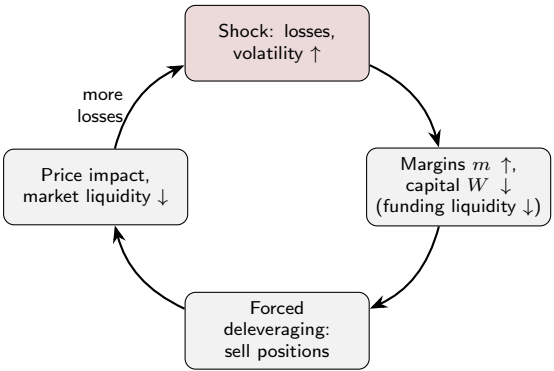
Channel 2 Formalised: Loss and Margin Spirals

Funding constraint of a leveraged fund (Brunnermeier & Pedersen 2009):

$$m_t \left| x_t \right| \leq W_t$$

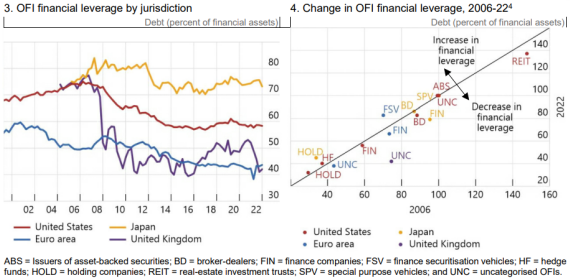
x_t = position, m_t = margin per unit, W_t = capital

- Losses: $W_t \downarrow$; volatility: $m_t \uparrow$
- $\Rightarrow$ the maximum position W_t / m_t falls on *both* counts
- Forced sales move prices $\Rightarrow$ further losses
- **Loss spiral + margin spiral**
- Liquidity providers turn into liquidity demanders

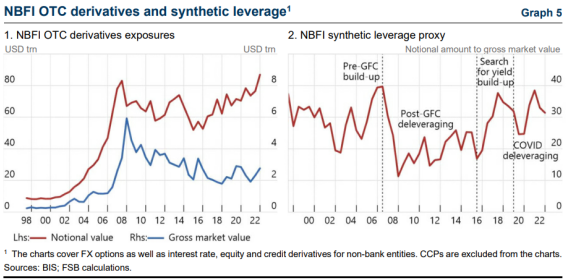


Financial vs. Synthetic Leverage Trends

Direct Financial Leverage



Synthetic Leverage (Derivatives)



Takeaway: Balance-sheet leverage of OFIs is below its pre-2008 level, especially in the UK and the US, but leverage has moved into derivatives: NBFI OTC notional reached a new high above $80 trillion in 2022.

Source: FSB (2023), The Financial Stability Implications of Leverage in Non-Bank Financial Intermediation, Graph 2 (panels 3–4) and Graph 5.

Channel 3: Cross-Border Spillovers & USD Funding

The global NBFI sector depends heavily on U.S. Dollar funding, regardless of its home base.

- **The FX Swap Nexus:** Non-US funds use short-term FX swaps to hedge USD assets. Off-balance sheet USD obligations exceed **$80 trillion** (end-June 2022), of which ≈$26 trillion are owed by non-banks outside the US (Borio, McCauley & McGuire, BIS QR Dec 2022).
- **Refinancing Risk:** In the "Higher-for-Longer" regime, duration-heavy portfolios face constant valuation pressure and refinancing "cliffs."
- **Global Contagion:** A US Hedge Fund failure directly hits its European G-SIB Prime Broker, restricting credit to local EU firms.

Channel 4: The Correlation Channel (Herding)

Market intermediation and portfolio choices are now extremely concentrated.

- **Herding:** High use of ETFs and algorithmic strategies means diverse players act as a single block during stress.
- **Plumbing Risk:** A small number of G-SIBs dominate Repo, Prime Brokerage, and Clearing.
- **Correlation $\rightarrow$ 1:** In a crisis, the "safe" and "risky" assets start moving together, breaking traditional diversification models.

  "If the central 'plumbing' (G-SIBs) retreats from market-making due to their own risk limits, liquidity vanishes for the entire system."

Synthesis: The Multiplier Effect

The Chain Reaction:

1. **Shock:** Interest rates rise or CRE defaults spike.
2. **Valuation:** Bond prices fall $\rightarrow$ Margin calls issued.
3. **Liquidity:** NBFIs dump blue-chips to meet calls.
4. **Contagion:** Banks face losses on credit lines and de-risk.
5. **Impact:** Credit crunch for the real economy.

Conclusion: NBFI risk is no longer "contained" outside the banking system; it is the primary driver of modern financial instability.

Case Study 1: March 2020 – The "Dash for Cash"

- **The Shock:** Pandemic uncertainty triggers a global flight to safety.
- **The Paradox:** Even US Treasuries (the ultimate safe asset) saw prices plummet and yields spike due to unprecedented NBFI selling pressure.
- **Transmission:**
  - **Open-ended funds** faced massive redemptions and sold liquid assets first.
  - **Hedge funds** unwinding "basis trades" (arbitrage between cash and futures) accelerated the fire-sales.
  - **Bank dealers** were constrained by post-GFC balance sheet regulations (SLR), preventing them from absorbing the selling volume.
- **Lesson:** In a systemic crisis, NBFI liquidity needs can easily overwhelm the banking sector's capacity to act as a market maker.

Case Study 1: March 2020 – The Dollar Funding Side

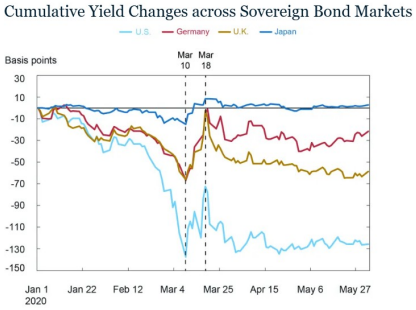
- **FX swap market:** the cross-currency basis widened sharply → synthetic dollar funding became very expensive for non-US banks, insurers and funds hedging USD assets
- **Policy response:**
  - 15 March: standing swap lines (BoC, BoE, BoJ, ECB, SNB) repriced to OIS + 25 bp, with longer maturities
  - 19 March: temporary lines for 9 more central banks (14 in total); FIMA repo facility for other official holders from 31 March
  - Peak usage USD 449 bn (week of 27 May 2020)
- **Outcome:** CIP deviations narrowed more for swap-line counterparties than for others (Bahaj & Reis 2022)

Lesson

The Fed acted as the global lender of last resort in dollars. Countries outside the swap-line network had to rely on their own FX reserves (see Channel 3).

Source: Yale Program on Financial Stability; BIS Bulletin No. 34 (2020).

Case Study 1: March 2020 – Yields

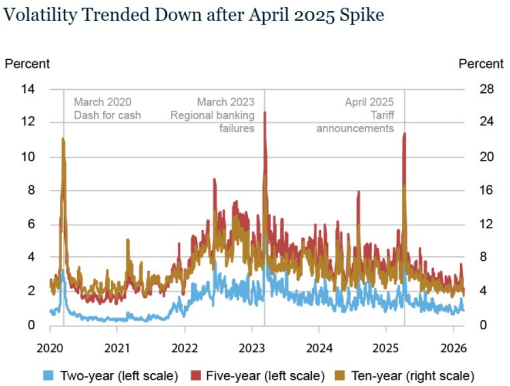


Source: Bloomberg L.P.
Notes: The chart displays the cumulative yield changes for ten-year sovereign bonds, starting on January 1, 2020. U.S., Germany, U.K., and Japan denote Treasury, bund, gilt, and Japanese government bond (JGB) securities, respectively.

Takeaway: Between 10 and 18 March 2020 ten-year yields rose 60–70 basis points in the US, the UK and Germany at the same time: investors sold the safest bonds for cash – the opposite of a flight to quality.

Source: Barone, Chaboud, Copeland, Kavoussi, Keane and Searls (2022), "The Global Dash for Cash in March 2020", Liberty Street Economics, 12 July; data: Bloomberg L.P.

Case Study 1: Treasury Market Volatility, 2020–2026



Takeaway: Treasury volatility has spiked three times since 2020 – the dash for cash, the 2023 bank failures and the April 2025 tariff shock – each time to similar heights, and each time it subsided within weeks.

Source: Dyer and Fleming (2026), "Treasury Market Liquidity Since April 2025", Liberty Street Economics, 2 April.

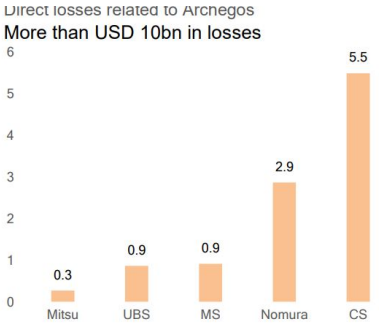
Case Study 2: Archegos (2021) – Synthetic Leverage Risks

- **Mechanism:** A family office used **Total Return Swaps** (TRS) with multiple global G-SIBs to build a concentrated \$50B+ exposure.
- **The Failure:** Because the leverage was synthetic (off-balance sheet), no single prime broker knew the true size of the client's total position.
- **Impact:** A margin call triggered a fire-sale, resulting in **\$10B+** in losses for banks (Credit Suisse $\approx$ \$5.5B, Nomura $\approx$ \$2.9B).
- **Race to exit:** PBs that sold their collateral first, in large blocks, lost least – a coordination problem among creditors.

Regulatory Lesson

Transparency in derivatives is a "blind spot." Prime brokers must assess **aggregated leverage** across the entire financial system, not just their own books.

Case Study 2: Archegos (2021) – Direct Losses

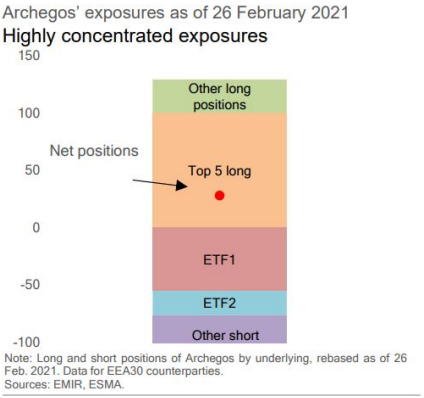


Note: Reported losses in USD bn as of April 2021. Mitsu: Mitsubishi UFG; MS: Morgan Stanley; CS: Credit Suisse,
Sources: Company disclosures, ESMA.

Takeaway: Archegos cost its prime brokers more than \$10 billion, over half of it at Credit Suisse (\$5.5 billion): each bank saw only its own swaps with Archegos, not the total position.

Source: ESMA (2022), Leverage and derivatives – the case of Archegos, TRV Risk Analysis, May; company disclosures.

Case Study 2: Archegos (2021) – Concentration Risk



Takeaway: Five stocks made up most of Archegos's long exposure, while its shorts were mainly ETFs – the book looked balanced in gross terms but was a concentrated bet on a handful of names.

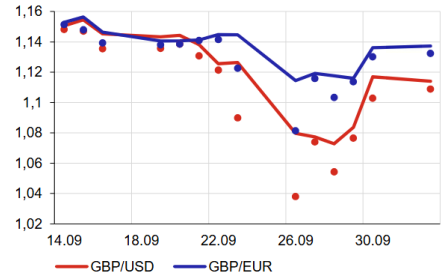
Source: ESMA (2022), Leverage and derivatives – the case of Archegos, TRV Risk Analysis, May; EMIR data (EEA30 counterparties).

Case Study 3: UK LDI Crisis (2022) – The Pension Fund Trap

- **Context:** UK pension funds used Liability-Driven Investment (LDI) strategies with **3x leverage** to match future liabilities.
- **Shock:** Rapid rise in Gilt yields (triggered by the 23 September 2022 "mini-budget"; long yields up by more than 100 bp within days) led to massive margin calls on LDI derivative positions.
- **The Death Spiral:** To raise cash for margin, funds sold Gilts $\rightarrow$ Gilt prices fell $\rightarrow$ yields rose $\rightarrow$ more margin calls.
- **Bank Intervention:** The Bank of England had to act as "Buyer of Last Resort" to prevent a systemic collapse of the pension sector (temporary purchases of long-dated gilts from 28 September, $\approx$ £19 bn).
- **Lesson:** Even "conservative" NBFIs (pension funds) can become systemic risks through leveraged hedging strategies.

LDI Crisis: Impact on Exchange rate

Graf 2 – Kurz GBP/EUR a GBP/USD
(jednotky měny za 1 GBP)

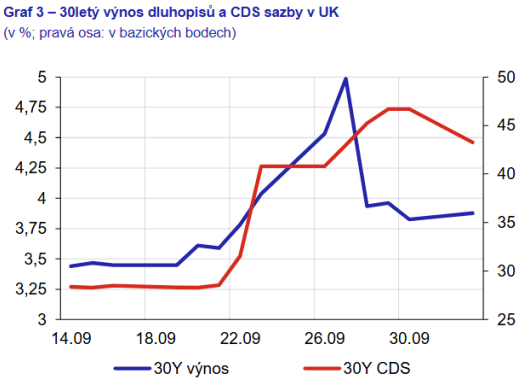


Zdroj: Refinitiv
Poznámka: Tečky označují zaznamenané denní minimum.

Takeaway: Sterling hit a record low against the dollar (≈1.03, 26 September 2022; dots = daily lows) while gilts sold off – currency and government bonds fell together, as in an emerging market under a credibility shock.

Source: Refinitiv; Časta, Siuda and Holub (2024), "Vše, co jste kdy chtěli vědět o investičních fondech (a báli jste se zeptat)", ČNB blog, 18 September, Graph 2.

LDI Crisis: Impact on Gilt Yields



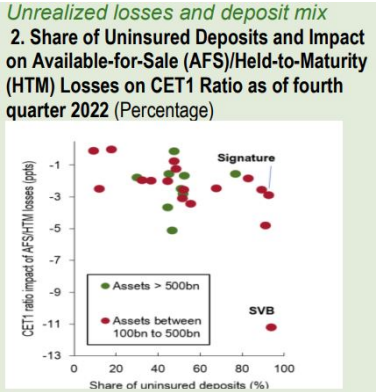
Takeaway: 30-year gilt yields jumped about 150 basis points in a week to 5% as LDI funds sold gilts to meet collateral calls, then fell about 100 basis points on 28 September, when the Bank of England started buying long gilts; UK CDS rose too.

Source: Refinitiv; Časta, Siuda and Holub (2024), "Vše, co jste kdy chtěli vědět o investičních fondech (a báli jste se zeptat)", ČNB blog, 18 September, Graph 3.

Case Study 4: SVB (2023) – The NBFI Deposit Connection

- **The Linkage:** Silicon Valley Bank (SVB) had a deposit base highly concentrated in **VC and Private Equity funds**.
- **Transmission:** As VC funding dried up, these NBFI clients withdrew deposits simultaneously.
- **The Digital Run:** Social media and digital banking accelerated the flight, with $42B in withdrawal requests in 24 hours.
- **Lesson:** Sectoral concentration in bank funding (relying on NBFIs) creates extreme vulnerability. **Correlation of depositor behavior** is a modern systemic risk.

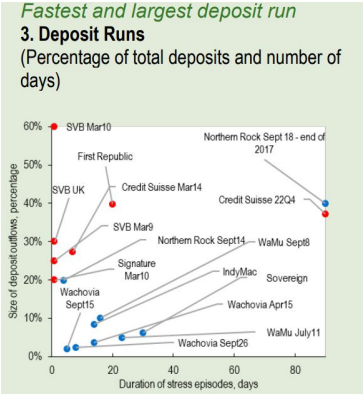
Case Study 4: SVB (2023) – HTM Losses



Takeaway: SVB was the outlier on both axes: over 90% of its deposits were uninsured, and unrealised securities losses would have cut its CET1 ratio by about 11 percentage points.

Source: IMF, Global Financial Stability Report, April 2023, Chapter 1.

Case Study 4: SVB (2023) – Bank Runs



Takeaway: SVB lost a quarter of its deposits on 9 March 2023 and faced requests for about 60% more the next day – runs that took weeks at Northern Rock or Washington Mutual happened within hours.

Source: IMF, Global Financial Stability Report, April 2023, Chapter 1.

Comparative Analysis: What have we learned?

Episode	Core Driver	Key NBFI Sector	Systemic Impact
March 2020	Market Liquidity	MMFs / Hedge Funds	Global Central Bank Intervention
Archegos	Hidden Leverage	Family Offices	Severe Bank Capital Losses
UK LDI	Margin Spirals	Pension Funds	Sovereign Bond Market Instability
SVB 2023	Funding Concentration	VC / PE Funds	Regional Banking Crisis

Key Takeaways for Students

- **Leverage is often hidden** until a price shock occurs.
- **Liquidity mismatch** is the primary "fuel" for systemic fire-sales.
- **Interconnectedness** means that a problem in a "small" fund can threaten G-SIBs.

Geopolitical Fragmentation as a Financial Risk

- **Capital Misallocation:** Cross-border flows are increasingly driven by geopolitical alignment ("friend-shoring") rather than fundamentals.
- **Shock Amplification:** Geopolitical events trigger rapid capital flight from emerging markets, straining local bank liquidity and NBFI sectors.
- **Inflationary Pressure:** Reshoring supply chains leads to "higher-for-longer" interest rates, putting permanent pressure on leveraged NBFIs.

Policy Implications: Three Lines of Defence

1. See the risks

Granular data on leverage and exposures, shared across borders (trade repositories, EMIR, SFTR); system-wide stress tests that cover banks, funds, insurers and CCPs together.

2. Make non-banks absorb their own shocks

- Liquidity management tools in open-ended funds: anti-dilution levies, swing pricing, notice periods – redemption terms that match asset liquidity.
- Margin and collateral preparedness; leverage limits where risks concentrate; independent valuation standards for private credit.

3. Make markets and backstops work under stress

Central clearing of government bonds and repo; central bank facilities for non-banks – priced, and conditional on resilience, to limit moral hazard.

What Has Been Agreed Internationally (2021–2025)

Area	Measure	Status
Money market funds	FSB policy proposals (Oct 2021); US SEC reform (Jul 2023): higher liquidity minimums, redemption gates removed, mandatory liquidity fees for institutional prime funds	Implementation varies by jurisdiction
Open-ended funds	FSB revised recommendations + IOSCO anti-dilution guidance (Dec 2023): redemption terms matched to asset liquidity; wider use of anti-dilution tools	FSB/IOSCO stocktake by end-2026
Margin and collateral calls	FSB: eight recommendations on non-banks' liquidity preparedness (Dec 2024)	Being implemented
Leverage	FSB final recommendations (Jul 2025): identify the risks, then act with a flexible menu of measures	Supervisory follow-up; data test case on sovereign bond markets
US Treasury market	SEC central clearing mandate (Dec 2023)	Cash trades by 31 Dec 2026, repo by 30 Jun 2027

The FSB says its NBFI work is now shifting from policymaking to assessing vulnerabilities, closing data gaps and evaluating reforms (July 2025).

Source: FSB (2021, 2023, 2024, 2025); IOSCO (2023); US SEC (2023, 2025).

Europe and the UK: From Lessons to Rules

European Union

- **AIFMD II** (Directive 2024/927), applies from 16 April 2026: open-ended AIFs must select at least two liquidity management tools; loan-originating funds face leverage caps of 175% (open-ended) and 300% (closed-ended) and retain 5% of loans they sell; enhanced reporting from April 2027
- **Eurosystem task force (May 2026)** asks for more: stronger AIFMD leverage powers, minimum haircuts in non-centrally cleared repo, a new ex ante liquidity tool (notice periods) for funds incl. UCITS, an EU system-wide stress test, ESMA coordination and top-up powers
- **European Commission:** consulted in 2024; targeted macroprudential actions for non-banks under consideration for 2026 (Market Integration and Supervision Package, Dec 2025)

United Kingdom (after the 2022 LDI crisis)

- **Minimum resilience for LDI funds:** buffers to withstand a yield shock of about 250 basis points (FPC recommendation, March 2023; implemented by The Pensions Regulator)
- **Contingent NBFI Repo Facility** (opened January 2025): the Bank of England lends cash against gilts to insurers, DB pension schemes and LDI funds – only in severe gilt market dysfunction

Sequence

Resilience first, backstop second: the facility complements the buffers, it does not replace them.

Source: Directive (EU) 2024/927; Eurosystem FSC (2026), *Strengthening the macroprudential lens in the regulation of NBFI*; European Commission (2025); Bank of England (2023, 2025).

Open Questions – and the Czech Angle

- **Backstops vs moral hazard:** with the Fed's Standing Repo Facility (2021) and the BoE's CNRF (2025), central banks now stand behind some non-banks in stress. Who pays for that insurance – eligibility, fees, haircuts, resilience conditions?
- **Entity vs activity:** the same activity (e.g. leveraged government bond trading) sits in banks, hedge funds and pension funds. Regulate the activity – margins, haircuts – rather than the label?
- **Data gaps:** private credit and offshore hedge funds remain largely outside EU reporting; many North American and Caribbean hedge funds do not report to EU authorities (ESRB and ECB 2026).

The Czech angle

The ČNB has run annual liquidity stress tests of investment funds since 2019 (pension companies' funds included since 2022), published in its *Financial Stability Report*. With funds projected to reach about a fifth of financial-sector assets, the redemption terms and liquidity tools of funds holding less liquid assets (e.g. real estate) are the key domestic question.

Discussion

Should a central bank lend to an LDI fund or a hedge fund in a crisis? On what terms?

Source: ČNB, investment fund sector stress tests; ESRB and ECB (2026); Federal Reserve (2021); Bank of England (2025).

A Taxonomy of Modern Financial Fragility

Systemic risk today is layered, not localized.

- **1. Micro-Level (Entity):** Liquidity mismatches, hidden leverage, and valuation opacity in individual funds.
- **2. Meso-Level (Network):** High concentration in prime brokerage and recursive collateral chains.
- **3. Macro-Level (Regime):** Higher-for-longer rates and USD funding dependencies.

Central Insight

The system is safer at the core (well-capitalized banks), but significantly more fragile at the edges (the NBFI "blind spots").

Conclusion: The New Financial Landscape

- **Unprecedented Integration:** NBFIs now hold **51% of global financial assets**. The system is no longer bank-centric.
- **Risk Migration:** Risk has not been eliminated; it has moved to less transparent sectors (Private Credit, Hedge Funds).
- **Persistent Linkages:** The banking sector (G-SIBs) is highly capitalized, but their deep linkages to NBFIs mean they remain exposed to off-balance-sheet contagion.
- **Future Tests:** Geopolitical fragmentation and structurally higher interest rates will rigorously test the resilience of this deeply integrated, highly leveraged ecosystem.

The system is more resilient in parts, but more interconnected everywhere.

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